Pendem Medhansh

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.48
Narayana Junior College Nallakunta	Class XII (SSC)	2023	96.60%
Kendriya Vidyalaya No. 1 Uppal	Class X (CBSE)	2021	94.33%

WORK EXPERIENCE

Software Development Engineer Intern (Full-Stack AI/ML) | Autonomic Solutions LLC, Chantilly, Virginia, USA
(Jan 2026 - Jul 2026)

- Contributed full-stack features to 5 of the 6 core services of the **Autonomic Computing Platform (ACP)**, a multi-cloud automation product for US federal clients, across frontend, backend, DevOps, and cloud layers.
- Built a cloud platform on **AWS** using **Terraform** modules, with **GitHub Actions CI/CD** pipelines to automate build, test, and deployment of demo applications.
- Developed **LLM-based agents** for application code generation with an interactive plan-refinement step, and for NIST/FISMA compliance scoring with downloadable PDF reports.
- Worked in a small team under the founder, shipping features end to end, from UI to infrastructure.

PROJECTS

Autonomic Computing Platform (ACP): Agentic AI for Cloud Infrastructure Automation | *TypeScript, Node.js, Google Vertex AI, Agent Development Kit (ADK), AWS*

- Built an **agentic code-generation pipeline** on **Vertex AI** using **Google's Agent Development Kit**, converting user requirements into a structured generation plan, letting the user iteratively refine the plan through conversation, then generating ready-to-deploy application code from the approved plan.
- Designed a **NIST/FISMA compliance-scoring agent**, where the LLM evaluates infrastructure scan results against Low/Moderate/High baselines and produces a compliance score with a generated PDF report.
- Built a **remediation agent** in the observability service, on a **MAPE-K control loop**, where the agent analyzes infrastructure health and triggers **one-click** automated self-healing.
- Worked across prompt design, agent orchestration with ADK, and the application layer integrating these agents into a production multi-cloud platform (AWS/Azure/GCP).

Conflict-Aware RAG System

- Built a **conflict-aware RAG system** using Gemini Flash and ChromaDB to answer enterprise policy queries despite contradictory information across documents.
- Designed a **custom retrieval-ranking algorithm** combining semantic similarity with role-based, policy-specificity, and recency boosts to select the authoritative document.
- Implemented **metadata-aware document processing and chunking**, capturing user roles, policy types, dates, and source information to enable contextual conflict resolution.
- Generated **transparent, citation-backed responses** containing the final ruling, supporting evidence, reasoning behind document selection, and explanations for conflicting sources.

Advanced Brain Tumor Segmentation & Classification Using Attention U-Net and Skip Connections

- **Implemented a custom Attention U-Net** from scratch to generate **precise segmentation masks** localizing brain tumors, including very small regions, achieving a final **Dice score of 79%.**
- **Researched and developed a novel hybrid loss function** combining **Dice loss** and **Binary Cross-Entropy** with a tunable hyperparameter = 0.8 to address **class imbalance** and enhance **mask boundary precision.**
- **Incorporated skip connections** in the CNN classifier to enable deeper training and feature reuse, boosting classification accuracy to **93%**, and deployed the complete solution as a **Flask-based web app** for real-time MRI scan analysis.
- **Tech stack:** TensorFlow, Flask, HTML, CSS, OpenCV (image processing)

SKILLS

- **Languages:** C, C++, Python, JavaScript, TypeScript, SQL
- **AI/ML:** LangChain, RAG, TensorFlow, Keras, PyTorch, Scikit-Learn, FAISS, Sentence-BERT, OpenCV
- **Core CS:** Data Structures & Algorithms, Object-Oriented Programming
- **Backend & Web:** Node.js, Express, Angular, Flask, REST APIs, HTML, CSS
- **Databases:** MySQL, PostgreSQL, Redis
- **Cloud & DevOps:** AWS, Docker, Kubernetes, Terraform, GitHub Actions, CI/CD
- **Data & Analytics:** Excel, Pandas, Matplotlib, Seaborn

POSITIONS OF RESPONSIBILITY

- Public Relations Head at **EBSB**, one of IITH's most vibrant cultural clubs, managing publicity and promotion of events.
- Inbound Coordinator at **International Relations Cell**. Managed international programs and maintained relations with Japanese universities. Handled various aspects of international relations effectively.
- Internship Coordinator at **OCS IITH**, managing communication between students and recruiters.
- Core member at **Rang De Manch**, the dramatics and speaking arts club of IITH, led multiple stage performances.
- Volunteer at **OCS IITH** in first year, assisted seniors in internship and placement season.